

Lecture 26

$$\# \quad Z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} \quad z_i \in \mathbb{C}^n$$

$z^T z$ is no longer the length because length can be negative then.

The new length is given by

$$\boxed{\bar{z}^T z} \Rightarrow (a_1^2 + b_1^2) + (a_2^2 + b_2^2) + \dots$$

$$|\bar{z}_1 z_1| = |z_1|^2$$

$$\# \quad Z^H = \bar{Z}^T \quad (H \Rightarrow \text{Hermitic symbol})$$

$$Z^H \rightarrow Z \text{ Hermitian}$$

Inner product now becomes $y^H x$

• Now, most of the times where we were doing just transpose, we take the hermitian.

• Symmetric matrices

$$A^T = A \quad \times \times \quad (\text{only for real})$$

$$\boxed{A^H = A}$$

✓✓

[for both complex]

$$\Rightarrow \begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix}$$

* Symmetric matrices with complex numbers are called Hermitian matrices.

↳ Perpendicular eigen vectors

$$Q = \begin{bmatrix} | & | & \dots \\ q_1 & q_2 & \dots \\ | & | & \dots \end{bmatrix}$$

$$q_i^T q_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

$$\boxed{Q^H Q = I}$$

orthogonal $\rightarrow$ unitary in
complex

Fourier Matrix

$$F_n = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & w & w^2 & \dots & w^{n-1} \\ 1 & w^2 & w^4 & \dots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \dots & w^{(n-1)^2} \end{bmatrix}$$

$$(F_n)_{ij} = \begin{cases} w^{ij} & i, j \in (0, n-1) \end{cases}$$

- Here w is a complex number
 $w^n = 1$

$$w = e^{i \frac{2\pi}{n}} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$$

For $n=4$

$$W = e^{i\pi/2}$$

$$W^4 = 1$$

$$W = i$$

$$i, i^2 = -1, i^3 = -i, i^4 = 1$$

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -i \\ 1 & -i & -1 & i \end{bmatrix}$$

$$\frac{1}{2}F_4 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$$

↓
now the columns are
orthonormal

So, to take inverse of orthonormal columns, we just transpose them.

$$F_4^H F_4 = I$$

$$F_4^H = F_4^{-1}$$

Fast Fourier Transform

Observation $\rightarrow$

$$(w_{2n})^2 = w_n$$

eg $w_6^2 = w_3$

$\rightarrow$ We can see this fact from the unit circle on the complex plane.

$$\# \quad [F_{64}] = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_{32} & 0 \\ 0 & F_{32} \end{bmatrix} \begin{bmatrix} 1 & 0 & \dots & 1 \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

$\swarrow$ Diagonal matrix

$\nwarrow$ P

takes even then odd components.

$$D = \begin{bmatrix} 1 & & & \\ & \omega & & \\ & & \omega^2 & \\ & & & \ddots \\ & & & & \omega^{31} \end{bmatrix}$$

⇒ Doing this we significantly reduce the number of calculations from $64^2 \rightarrow 2 \cdot (32^2) + \text{fixups}$

⇒ Now we keep dividing even the F_{32} further into F_{16}

$$F_{32} = \left[\begin{array}{c|c} I & D \\ \hline I & -D \end{array} \right] \left[\begin{array}{cc} F_{16} & 0 \\ 0 & F_{16} \end{array} \right] \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right]$$

$$F_{64} = \left[\begin{array}{cc} I & D \\ I & -D \end{array} \right] \left[\begin{array}{cc|cc} I & D_1 & I & D_1 \\ \hline I & -D_1 & I & -D_1 \end{array} \right] \left[\begin{array}{cc|cc} F_{16} & 0 & 0 & 0 \\ 0 & F_{16} & 0 & 0 \\ \hline 0 & 0 & F_{16} & 0 \\ 0 & 0 & 0 & F_{16} \end{array} \right] \dots$$

$$\Rightarrow 2 \cdot 32^2 + 32$$

↓ After breaking it into 16

$$2 \cdot (2 \cdot (16^2) + 16) + 32$$

→ We can keep breaking the problem down further & further.